Author Response

Anti-correlated Noise Provably Ensures Stable Convergence to Flat Minima

TABLE R.1. Test accuracy (%) on the real-world datasets CIFAR-10N/100N, where Nr denotes the estimated label-noise rate for each datasets. For CIFAR-10N, we selected Rand1, Aggregate, and Worst noise types to account for various real-world noise rates. Results are reported as the mean $\pm$ standard deviation over the last 10 epochs of a 150-epoch training process across 5 independent trials with different random seeds. All models use ResNet-34 and were trained from scratch using SGD with a batch size of 128, an initial learning rate of 0.02, a momentum of 0.9, and a weight decay of 1×10^{-3}. The learning rate decays by a factor of 0.1 at epochs 40, 80 (CIFAR-10N) and epochs 80, 120 (CIFAR-100N), following (Liu et al. 2022). Anti-SGD($\tau = \eta$), PSGD (i.e., SGD with white noise injection), and cANI share the same perturbation strength for each dataset, set to 0.1 for CIFAR-10N and 0.03 for CIFAR-100N. For cANI, the parameter τ is set to 0.05 and 0.5 for CIFAR-10N and CIFAR-100N, respectively. The equivalence between Anti-SGD and cANI no longer holds due to the learning rate scheduling.

	CIFAR-10N			CIFAR-100N
	Rand1 (Nr $\sim$ 0.09)	Aggregate (Nr $\sim$ 0.17)	Worst(Nr $\sim$ 0.4)	Noisy(Nr $\sim$ 0.4)
SGD	84.51 ± 0.36	89.83 ± 0.33	66.38 ± 0.52	54.28 ± 0.40
PSGD	40.74 ± 1.35	43.15 ± 0.54	33.98 ± 1.06	31.66 ± 0.23
Anti-SGD	81.83 ± 0.21	88.30 ± 0.22	62.79 ± 0.40	54.29 ± 0.27
cANI	$\mathbf{88.67 \pm 0.21}$	$\mathbf{90.34 \pm 0.16}$	$\mathbf{81.50 \pm 0.32}$	$\mathbf{58.38 \pm 0.32}$

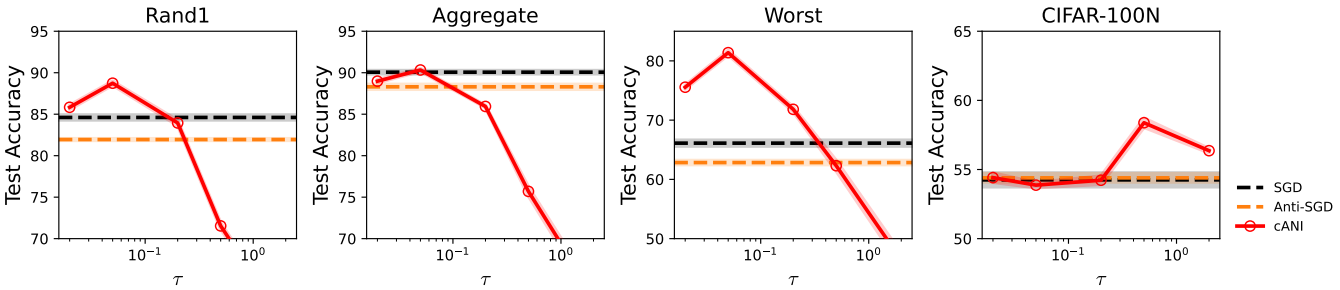

Figure R.1. **Non-monotonic τ dependence of cANI in CIFAR-10N/100N:** Test accuracy(%) of the cANI as a function of τ. We used $\sigma_A = 0.1$ for CIFAR-10N datasets (i.e., Rand1, Aggregate and Worst) and $\sigma_A = 0.03$ for CIFAR-100N. Other experimental settings are identical to the Table R.1. Shaded regions denote the standard deviation across 3 independent trials with different random seeds.

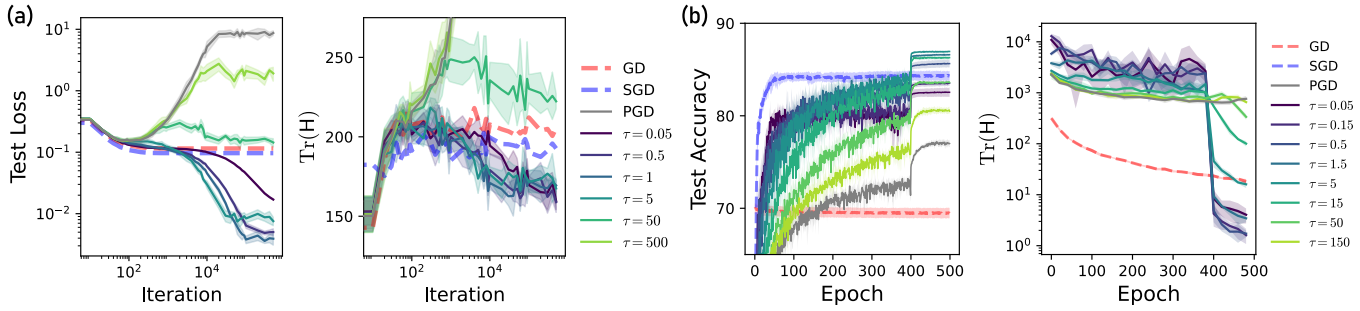

Figure R.2. **Revised Fig. 4 including PGD results:** Results of GD with anti-correlated noise on different tasks. (a) Matrix sensing task: Test loss (left) and trace of the Hessian Tr(H) (right) versus iteration for different τ with $\sigma_A = 0.01$. (b) Image classification on CIFAR-10: Test accuracy (left) and Tr(H) (right) versus epochs for different τ with $\sigma_A = 0.03$. Dashed lines represent the baselines: standard GD (red) and SGD (blue). The gray solid lines indicate the PGD baselines with matched noise strength and the results for $\tau = 0.05$ correspond to the Anti-PGD baselines (i.e., $\tau = \eta$). Shaded regions denote the standard deviation across 5 independent trials with different random seeds.

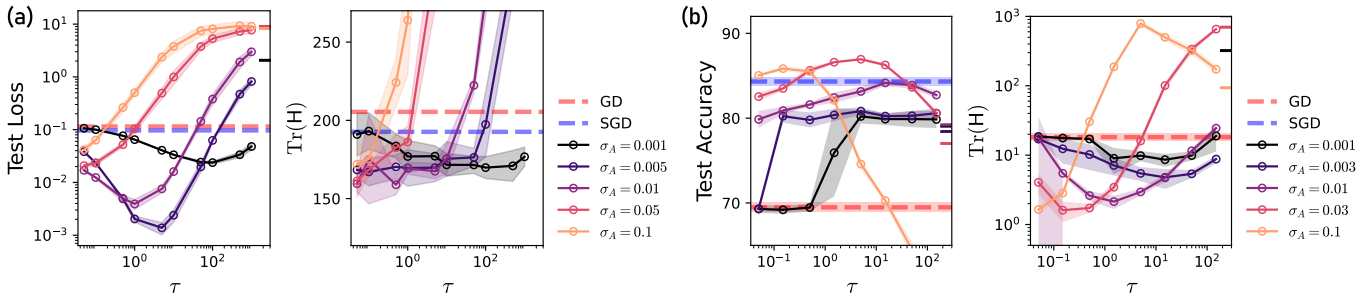

Figure R.3. **Revised Fig. 5 including PGD results:** Results of GD with anti-correlated noise on different tasks. (a) Matrix sensing task: Test loss (left) and trace of the Hessian Tr(H) (right) as a function of τ. (b) Image classification on CIFAR-10: Test accuracy (left) and Tr(H) (right) versus τ. Horizontal dashed lines represent the baselines: standard GD (red) and SGD (blue). The PGD baselines with matched noise strength are marked by short horizontal lines on the rightmost axis, representing the limit as $\tau \to \infty$. The Anti-PGD baselines correspond to the results at the left-most τ, representing the correspondence at $\tau = \eta$. Solid lines with markers indicate the results for varying σ_A, and shaded regions denote the standard deviation across 5 independent trials with different random seeds.

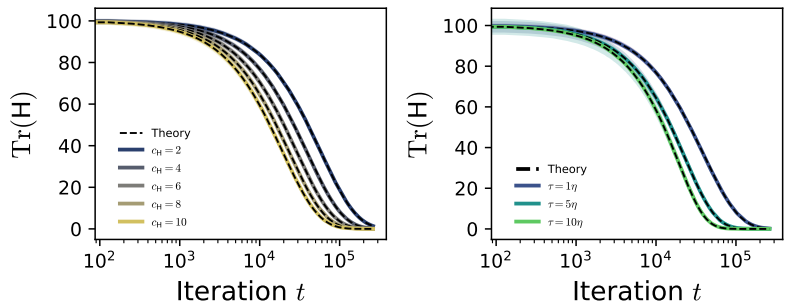

Figure R.4. **Anisotropic Hessian-aligned noise on the widening valley:** Evolution of the Hessian trace $\mathrm{Tr}(\mathsf{H})$ over iteration t for different c_{H} (left) and varying τ/η (right). Here, the injected noise is anisotropic and aligned with the Hessian, with its covariance given by $\eta c_{\mathsf{H}}|H|$. Solid colored lines denote the ensemble-average of $\mathrm{Tr}(\mathsf{H})$ and the shaded regions denotes the standard deviation of $\mathrm{Tr}(\mathsf{H})$, while black dashed lines denote the theoretical predictions. Theoretical paths follow the effective loss given by $\tilde{\mathcal{L}} = \mathcal{L} + \sigma_A^2 \mathcal{R}_\tau + c_{\mathsf{H}} \mathcal{R}_{\mathsf{H}}$, where the implicit regularizer from Hessian-aligned noise is given by $\mathcal{R}_{\mathsf{H}} = -(1/2)\mathrm{Tr}\left[\ln(\mathsf{I}_d - (\eta/2)\mathsf{H})\right]$. The simulations are conducted at the same initial points $\theta_f = 10, \boldsymbol{\theta}_s = \mathbf{0}$, with $\eta = 0.002$, $d_s = 1$,$\alpha_\tau = \sqrt{1 - \eta/2\tau}$ and tracked for every 2^7 iterations. We used $\sigma_A^2 = 2, \tau = \eta$ for the left figure and $c_H = \sigma_A^2 = 2$ for the right figure.

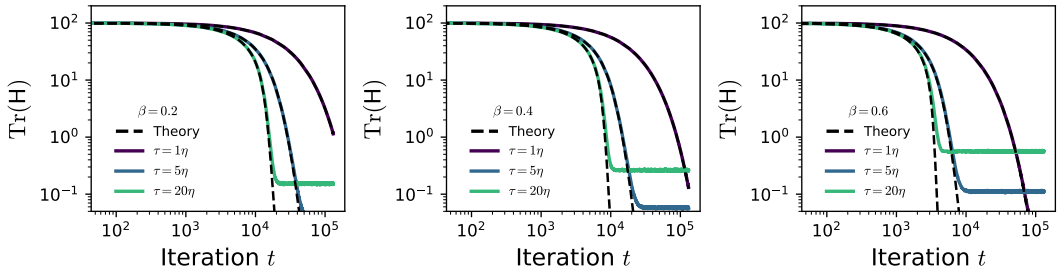

Figure R.5. **cANI on GD with momentum β on the widening valley:** Evolution of the Hessian trace $\mathrm{Tr}(\mathsf{H})$ over iterations t across τ with momentum $\beta = 0.2$ (left), $\beta = 0.4$ (middle), and $\beta = 0.6$ (right). Solid colored lines denote the ensemble average of $\mathrm{Tr}(\mathsf{H})$ and the shaded regions denote the standard deviation of $\mathrm{Tr}(\mathsf{H})$, while black dashed lines denote the theoretical predictions. Theoretical paths follow the effective loss given by $\tilde{\mathcal{L}} = \mathcal{L} + \sigma_A^2 \mathcal{R}_{\tau,\beta}$, where the implicit regularizer with momentum is given by $\mathcal{R}_{\tau,\beta} = \frac{1+\beta(1+\frac{\eta}{\tau})}{4(1-\beta)(1-\frac{\eta}{2\tau})(1+\beta(1-\frac{\eta}{\tau}))} \mathrm{Tr}\left[\ln\left(\mathsf{I}_d + (\tau - \eta)\left(\mathsf{H} - \frac{\beta}{\tau}\right)\right) - \ln\left(\mathsf{I}_d - \frac{\eta}{2(1+\beta)}\mathsf{H}\right)\right]$. The simulations are conducted from the same initial points $\theta_f = 10, \boldsymbol{\theta}_s = \mathbf{0}$, with $\eta = 0.05$, $\sigma_A = 1$, $d_s = 1$, and $\alpha_\tau = \sqrt{1 - \frac{\eta}{2\tau}}$, and are tracked every 2^6 iterations.

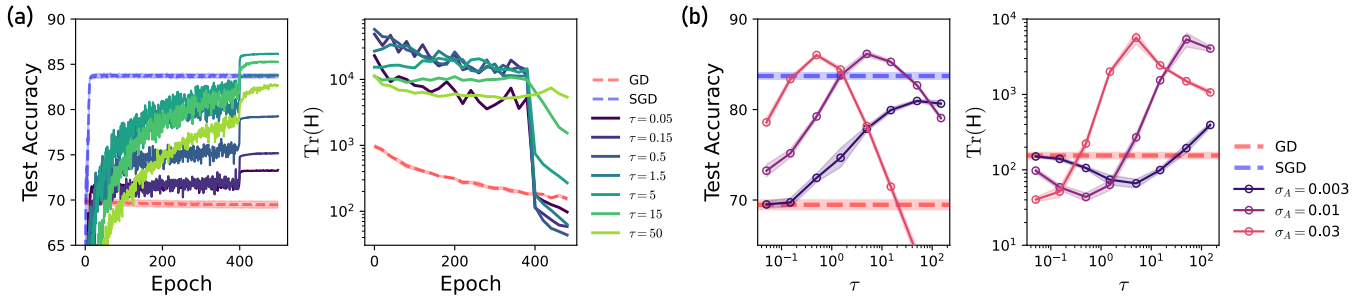

Figure R.6. Results of GD *without* momentum with anti-correlated noise for image classification task on CIFAR-10. (a) Test accuracy (left) and $\mathrm{Tr}(\mathsf{H})$ (right) versus epoch for different τ (solid lines) with $\sigma_A = 0.01$. (b) Test accuracy (left) and $\mathrm{Tr}(\mathsf{H})$ (right) versus τ for varying σ_A (solid lines with markers). Dashed lines represent the baselines: standard GD (red) and SGD (blue). Shaded regions denote the standard deviation across 5 independent trials with different random seeds. Other experimental details are identical to the Fig 5.(b) in the main article.

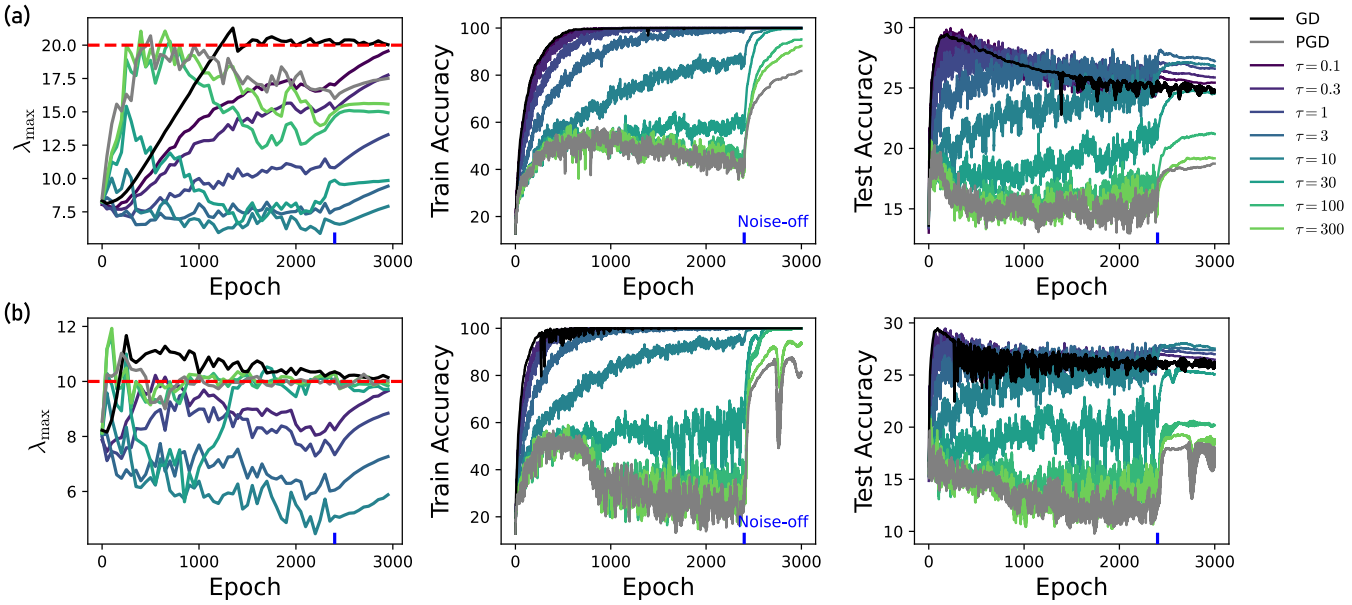

Figure R.7. **Edge-of-Stability (EoS) results:** Evolution of the top eigenvalue $\lambda_{\max}$ (left), training accuracy (middle), and test accuracy (right) over epochs with learning rates (a) $\eta = 0.1$ and (b) $\eta = 0.2$. Colored lines represent the results of GD with cANI for varying τ with $\sigma_A = 0.01$. The horizontal red dashed lines indicate the theoretical stability threshold at $2/\eta$. Solid black and gray lines represent standard GD and PGD with matched noise strength, respectively. The blue tick mark indicates the point where the injected noise is turned off (Noise-off) for the final 20% of training. Experiments were performed on a 500-sample subset of CIFAR-10 with the fully-connected ReLU network and the MSE loss, following the setup of (Cohen et al., 2022) but with a reduced dataset size.

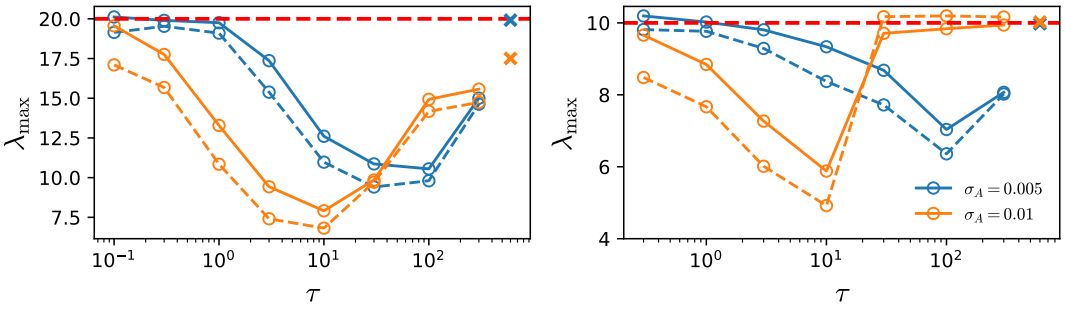

Figure R.8. **Edge-of-Stability (EoS), Comparison of $\lambda_{\max}$:** The top eigenvalue $\lambda_{\max}$ as a function of τ for learning rates $\eta = 0.1$ (left) and $\eta = 0.2$ (right) with varying σ_A. Solid lines indicate the results at the final training epoch, while dashed lines represent the results at the exact point of noise removal (Noise-off). The horizontal red dashed lines denote the theoretical stability threshold at $2/\eta$. The cross markers on the far right indicate the PGD baselines, representing the limit $\tau \to \infty$.

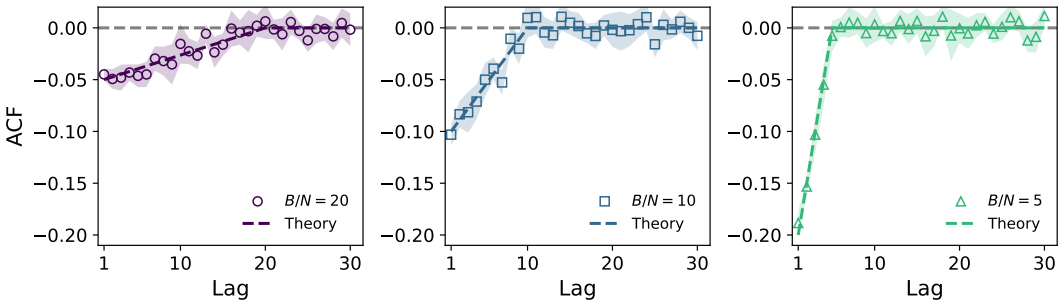

Figure R.9. **ACF of mini-batch noise:** Auto-correlation function (ACF) of mini-batch gradient noise versus lag with $B/N = 20$ (left), $B/N = 10$ (middle), and $B/N = 5$ (right), where B is the batch size and N is the dataset size. To estimate the ACF, the gradient noise was first projected onto random directions, and the ACF was then computed from the resulting scalar signals. Symbols indicate the mean experimental ACF over 10 random projections, shaded regions indicate one standard deviation across the 10 projected ACFs, and dashed lines indicate the theoretical predictions. Experiments were performed on a 500-sample subset of CIFAR-10 with ResNet-18.

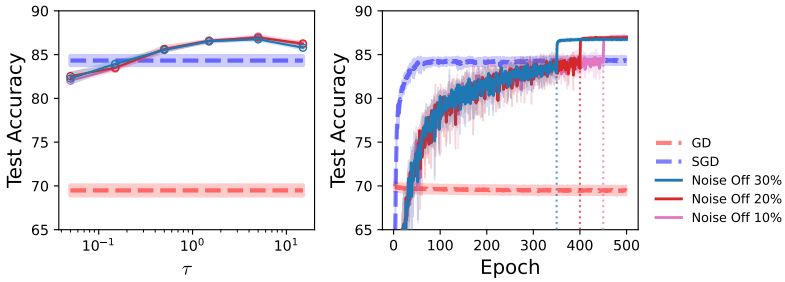

Figure R.10. **Robustness of noise-off schedule:** Test accuracy of cANI, GD and SGD across τ (left) and during training (right). In the left panel, Test accuracy is averaged over last 10 epochs. As in the Fig. 5(b) of the main article, ResNet18 was pretrained for 75 epochs with a batch size 2048, learning rate 0.05, momentum 0.9, and no weight decay. After the initialization, the anti-correlated noise with $\sigma_A = 0.03$ is applied, and turned off at 350 (70%), 400 (80%) and 450 (90%) epochs. For SGD, we changed batch size by 128 after the initialization. We used $\tau = 5$ for cANI in the right panel. The shaded regions denote the standard deviation over 4 different runs, and the dotted vertical lines in the right panel denotes the point when the noise injector is removed.

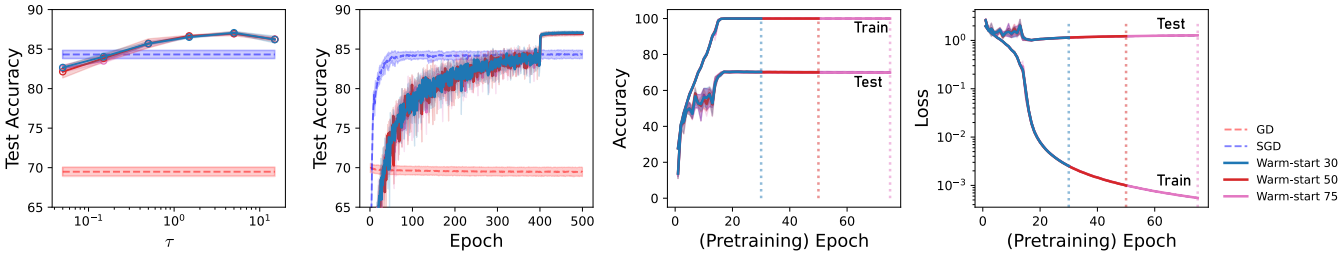

Figure R.11. **Robustness of warm-start schedule:** Test accuracy(averaged over last 10 epochs) of cANI, GD and SGD across τ (first) and during training (second) for different pretraining epochs. The train/test accuracy during pretraining are shown in third panel, and the train/test loss are shown in fourth panel. We used $\sigma_A = 0.03, \tau = 5$ for cANI, and turned off at 400 (80%) epochs, while other experimental setups are identical to Fig. R.10 . The shaded regions denote the standard deviation over 4 different runs, and the dotted vertical lines in third and fourth panels denote the point when the pretraining is over.

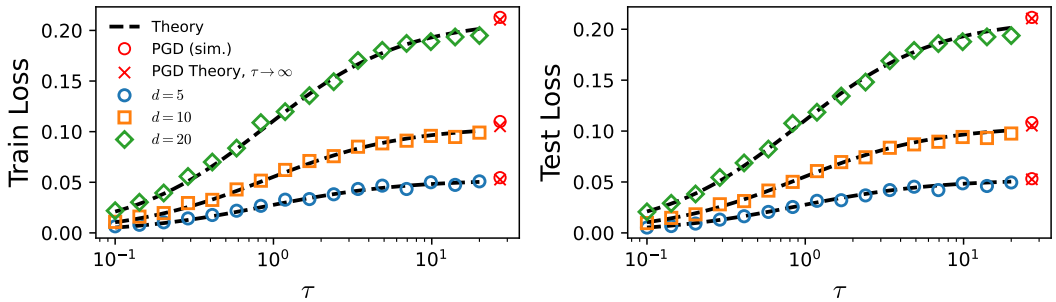

Figure R.12. **Explicit train/test-loss calculation in linear regression:** Ensemble-averaged train (left) and noiseless test (right) losses for different dimensions d as functions of τ on a logarithmic scale. We consider a linear-regression model with $\boldsymbol{x}_i, \boldsymbol{u}, \boldsymbol{w} \in \mathbb{R}^d$ and $y_i = \boldsymbol{u}^\top \boldsymbol{x}_i + \epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$, and fit $\boldsymbol{w}$ using the MSE loss. Colored markers (blue circles, orange squares, and green diamonds for $d = 5, 10, 20$, respectively) denote numerical averages, and black dashed lines denote the closed-form theory. Red markers on the far right denote the PGD baselines, representing the limit $\tau \to \infty$, with circles and crosses corresponding to simulation and theory, respectively.